
OBJECTIVES

- Understand and Master the Following UML Diagrams:
 - Use Case
 - Class Diagram

TEAM EXERCISES

Task / Exercise C0 (Introductory Questions) [10 minutes + 5 minutes discussion] :

Answer the questions briefly, Teams must assigned a student to write their answers:

- For a use case diagram:
 - The actor must be an end-user (humans) ?
 - What's the difference between include and extend
- In a class diagram :
 - What's the difference between Aggregation and Composition by giving examples
 - Why Association Classes cannot be replaced by just adding attributes ?

TEAM EXERCISES

Task / Exercise C1 (Use Case Diagram):

Draw the full UML Use Case Diagram for the "Quick Cup" Coffee Ordering App (seen in previous TD). Ensure the following elements are correctly modeled:

- Use of <<include>>: Identify a common sub-function that is mandatory and executed by multiple primary Use Cases (e.g., part of both Place Order and Customize Drink). Model this common function as a separate Use Case and connect it to its callers using the <<include>> relationship. Justify your choice.
- Use of <<extend>>: Propose an optional, non-core feature for the system (e.g., Apply Loyalty Points). Model this as an extending Use Case, clearly specifying the base Use Case and the Extension Point where it can be conditionally activated. Justify your choice.

Task / Exercise C2 (Class Diagram): Draw the Class-diagram for the following statement : *"There are two kinds of participants, namely members and guests. Guests are invited by members at a given date".*

Task / Exercise C3 (Use Case) :

Teams should split themselves into two sub-teams where each one should work on a case:

1. Research project submission System

A University Research Center manages a digital platform that supports the entire lifecycle of research project submissions. A faculty member can prepare a new project proposal by uploading required documents, adding collaborators from inside or outside the university, and specifying the budget and expected duration. Once submitted, the proposal is automatically forwarded to the Ethics Committee, which evaluates ethical compliance and may request clarifications. After the ethical approval, the proposal moves to the Scientific Council, where each reviewer can assign a score, attach comments, and either endorse or reject the proposal. If the Scientific Council endorses it, the platform notifies the Research Office, which then allocates the preliminary budget and assigns a financial officer in charge of monitoring the project. External collaborators cannot access the platform directly but receive email notifications when the faculty member adds them, inviting them to confirm their participation through a temporary access link. The system keeps a full history of all evaluations, comments, decisions, budget allocations, and collaborator confirmations. A reporting service periodically generates summary reports for the university management, showing statistics on proposals, success rates, and budget distribution. An authentication service and a file storage service are integrated into the platform, and the financial monitoring operations integrate with the university's Accounting System.

Draw the Use Case Diagram for the research project submission system

2. Educational submission system (Like Google Classroom)

- Every course in the system has lecturers assigned to it. This is done by one of the course administrators, who is also a lecturer. As part of a course, lecturers may create tasks and assess papers submitted by students. Therefore, the lecturers award points and give feedback.
- The course administrator defines which lecturer assesses which papers. At the end of the course, the course administrator also arranges for certificates to be issued. A student's grade is calculated based on the total number of points achieved for the submissions handed in.
- Students can take courses and upload papers.
- All users—students and lecturers—can manage their user data, view the courses and the tasks set for the courses (provided the respective user is involved in the course), and view submitted papers as well as grade points. However, students can only view their own papers and the related grades. Lecturers can only view the papers assigned to them and the grades they have given. The course administrator has access rights for all data.
- A course is created and deleted by an administrator.
- When a course is created, at least one administrator must be assigned to it. Further course administrators can be assigned at a later point in

time or assignments to courses can be deleted. The administrator can also delete whole courses.

- *Information about users and administrators is automatically transferred from another system. Therefore, functions that allow the creation of user data are not necessary.*
- *All the system functions can only be used by persons who are logged in.*

Draw the Class Diagram from the previous description.

Task / Exercise P1

Your team has been asked to develop an information system for a small hospital. During the initial meeting, the hospital director explained how patient information is currently managed and provided you with a sample document in French language, called the "Dossier du patient"

This document is used to record essential data for each hospitalized patient, including the duration of each stay and the treatments administered. Each file can capture information for up to three hospital stays, but in practice, a patient may be admitted more than three times. When this happens, the staff creates additional files for the same patient.

From the director's explanations and the structure of the document, you learn the following:

- Every patient has one primary physician (their family doctor). Only one doctor acts in this role for any given patient.
- During a hospital stay, a patient may also be seen by consulting physicians. Several different doctors may intervene in this capacity during a single stay.
- Additionally, multiple doctors may provide specific treatments to the patient during a stay, and a stay may involve several treatments administered by different doctors.

Your team is asked to analyze the provided document and the director's description in order to:

- **Identify the system's functional requirements.**
- **Determine the actors and use cases.**
- **Produce the corresponding UML diagrams, including at least:**
 - **A Use Case Diagram**
 - **A Class Diagram**

Your analysis should be based strictly on the information provided; do not introduce assumptions not supported by the document or the director's explanation.

Dossier du patient

No dossier ➤ 2345678 **Adresse** ➤ 205, Potvin
 Lévis, QC, G6V 8R9
Nom, Prénom ➤ Lebel, Anne **Nom mère** ➤ Alice Nault
Date de naissance ➤ 25/03/60 **Médecin traitant** ➤ 344234
 Pierre Manseau

Séjour 1

Détails séjour	Traitements				Médecins consultés
	Type	Date	Durée	Médecin	
Motif hospitalisation:	Défibrillation	07/11/05	5 min.	678113	07/11/05
☒ Urgence	Angiographie	07/11/05	1 h.	713818	756454
☐ Chirurgie d'un jour	Pontages aor.	09/11/05	4 h.	781813	Sophie Matte
☐ Sur prescription					
☐ Autre					
Admission ➤ 07/11/05					09/11/05
Congé ➤ 14/11/05					809456
Diagnostic ➤ Infarctus du myocarde					Paul Auger

Séjour 2

Détails séjour	Traitements				Médecins consultés
	Type	Date	Durée	Médecin	
Motif hospitalisation:	Polypectomie	10/12/05	1 h.	785843	10/12/05
☐ Urgence					454756
☒ Chirurgie d'un jour					Yves Dion
☐ Sur prescription					
☐ Autre					
Admission ➤ 10/12/05					
Congé ➤ 10/12/05					
Diagnostic ➤ Polypes rectaux					

Séjour 3

Détails séjour	Traitements				Médecins consultés
	Type	Date	Durée	Médecin	
Motif hospitalisation:					
☐ Urgence					
☐ Chirurgie d'un jour					
☐ Sur prescription					
☐ Autre					
Admission ➤					
Congé ➤					
Diagnostic ➤					